

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA
FANIDAN
TEST TOPSHIRIQLARI KITOB

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Milliymath_sertifikat**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

– yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.

– ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'inishda yozing.

– moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!

28.02.2026. 1-smena.

1. Hisoblang:

$$\underbrace{666 \dots 6}_{50 \text{ ta}} + \underbrace{555 \dots 5}_{50 \text{ ta}} .$$

- A) $1 \underbrace{22 \dots 2}_{49 \text{ ta}} 1$
B) $1 \underbrace{22 \dots 2}_{48 \text{ ta}} 1$
C) $11 \underbrace{22 \dots 2}_{49 \text{ ta}} 1$
D) $1 \underbrace{22 \dots 2}_{50 \text{ ta}} 1$

2. Hisoblang:

$$\left(2, 28 - 4\frac{7}{20} \right) : 2\frac{1}{4} - \left(\frac{1}{2} - 1\frac{7}{8} \right) .$$

- A) $\frac{71}{100}$
B) $\frac{71}{200}$
C) $\frac{91}{200}$
D) $\frac{91}{100}$

3. Velosipedchi va motosiklchi bir-biriga qarab yo'lga chiqdi. Ular orasidagi masofa 275 km. Velosipedchi tezligi 20 km/h, motosiklchini tezligi 50 km/h. Agar velosipedchi 15 minut dam olgan bo'lsa, motosiklchi velosipedchi bilan uchrashguncha necha km masofa bosib o'tadi?

- A) 100
B) 150
C) 175
D) 200

4. Katetlari x, y va gipotenuzasi $3\sqrt{5}$ ga teng to'g'ri burchakli uchburchak uchun. x $133\frac{1}{3}\%$ ga va y $16\frac{2}{3}\%$ ga oshirilganda yig'indisi 14 ga teng bo'lsa, uchburchak yuzini toping.

- A) 18
B) 9
C) 12
D) 15

5. Hisoblang:

$$\left(\frac{2}{3}\right)^{-3} \cdot (1,875)^{-1}.$$

- A) $1\frac{3}{4}$
- B) 1,8
- C) $1\frac{5}{8}$
- D) 1,5

6. Hisoblang:

$$\sqrt{\frac{666666^2}{12345654321}}.$$

- A) 3
- B) 12
- C) 6
- D) 9

7. Hisoblang:

$$\sqrt[6]{7 - \sqrt{40}} \cdot \sqrt[3]{\sqrt{2} + \sqrt{5}} \cdot \sqrt[3]{9}.$$

- A) 2
- B) 3
- C) 4
- D) $3\sqrt{2}$

8. Arifmetik progressiyaning hadlari uchun $a_1 + a_2 = a_3$ va $a_3 \cdot a_1 \cdot a_2 = 384$ bo'lsa, S_{10} ni toping.

- A) 110
- B) 200
- C) 210
- D) 220

9. Kamayuvchi geometrik progressiyaning hadlari uchun $b_1 + b_2 + b_3 = 19$ va $b_1^2 + b_2^2 + b_3^2 = 133$ bo'lsa, S_5 ni toping.

- A) $23\frac{4}{9}$
- B) $23\frac{2}{9}$
- C) $23\frac{1}{9}$
- D) $22\frac{7}{9}$

10. $\frac{x}{y} = 4$ va $y \neq -\frac{3}{5}$ bo'lsa, $\frac{x+y+3}{x+6y+6}$ ning qiymatini toping.

A) $\frac{1}{3}$

B) 1

C) $\frac{1}{2}$

D) 2

11. Soddashtiring:

$$\frac{\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{xy}\right)(x+y-1)}{\frac{1}{x^2} + \frac{1}{y^2} + \frac{2}{xy} - \frac{1}{x^2y^2}} .$$

A) $\frac{1}{xy}$

B) xy

C) $\frac{1}{x^2y}$

D) 1

12. $x \in \left(0; \frac{\pi}{2}\right)$ uchun $\cos x = \frac{3}{5}$ bo'lsa, $a = \sin x$, $b = \operatorname{tg} x$ va $c = \operatorname{ctg} x$ sonlarini o'sish tartibida joylashtiring.

A) $a < b < c$

B) $c < a < b$

C) $c < b < a$

D) $b < a < c$

13. Agar $\alpha, \beta \in (0; 90^\circ)$ $\cos \alpha + \sqrt{3} \cos \beta = 0$ va $2\alpha + \beta = 180^\circ$ bo'lsa, $|\alpha - \beta|$ ning qiymatini toping.

A) 30°

B) 60°

C) 90°

D) 45°

14. Tenglamaning haqiqiy ildizlari yig'indisini toping:

$$4^{x^2-x} - 17 \cdot 2^{x^2-x+2} + 256 = 0 .$$

A) 2

B) 3

C) -2

D) 0

15. Tenglamaning haqiqiy ildizlari sonini toping:

$$x^4 - \frac{2027!}{2026! + 2025!} \cdot x^2 + 2025 = 0 .$$

- A) 3 ta
- B) 2 ta
- C) 5 ta
- D) 4 ta

16. Tenglamaning butun ildizlari sonini toping:

$$x^4 - \sqrt{|x| - \sqrt{2025}} = 2025^2 .$$

- A) 3 ta
- B) 2 ta
- C) 5 ta
- D) 4 ta

17. Tenglamaning haqiqiy ildizlari ko'paytmasini toping:

$$\frac{2}{3} - \log_x 3 = \log_3 2 \cdot \log_{32} x - \frac{2}{3} \log_{\sqrt{x}} 3 .$$

- A) $\sqrt[3]{3}$
- B) $9\sqrt[3]{3}$
- C) $27\sqrt[3]{3}$
- D) 27

18. Tengsizlikning butun yechimlari sonini toping:

$$\sqrt{x^2 - 4x + 4} < 5 \cdot \sqrt{2 - x} + 6 .$$

- A) 33 ta
- B) 32 ta
- C) 34 ta
- D) 36 ta

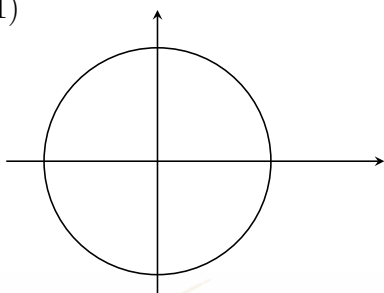
19. Tengsizlikni butun yechimlari yig'indisini toping:

$$\frac{x^2 - 7|x| + 10}{x^2 - 6x + 9} < 0 .$$

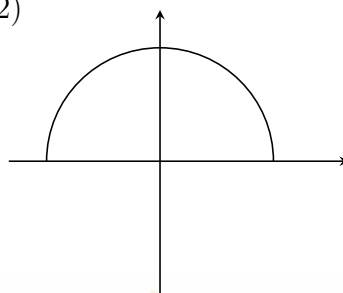
- A) -7
- B) -4
- C) -5
- D) -3

20. Quyidagilardan qaysi biri funksiya.

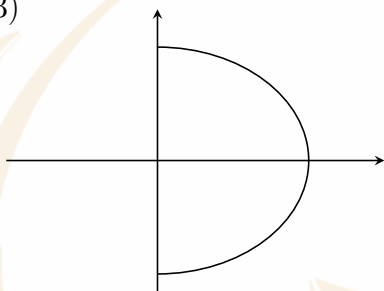
1)



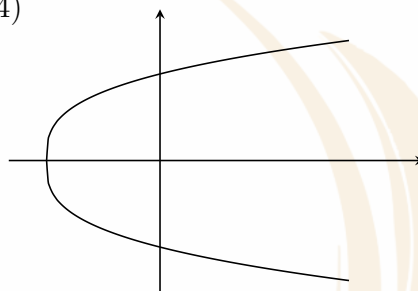
2)



3)



4)



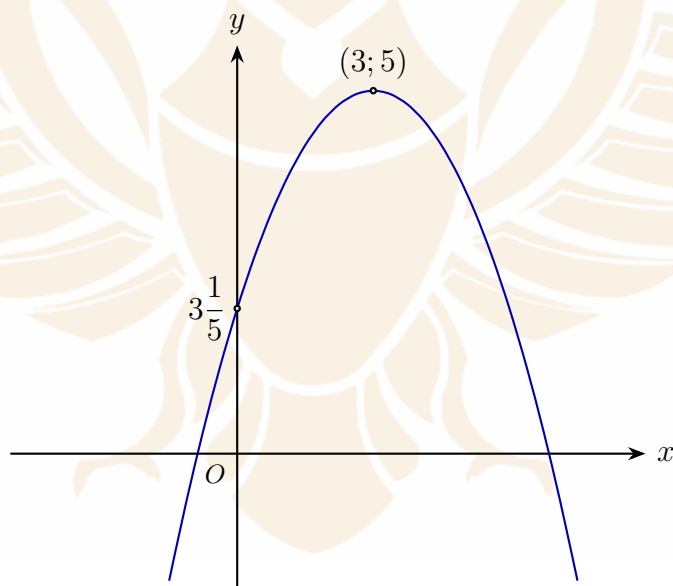
A) 1

B) 2

C) 3

D) 4

21. $f(x) = ax^2 + bx + c$ parabolaning grafigi quyida tasvirlangan. Shunga ko'ra, $f(18)$ ni toping.



A) -40

B) -20

C) 20

D) 40

22. Agar $3 \cdot (3x + 4)^{20} = C_0 x^{20} + C_1 x^{19} + \dots + C_{19} x + C_{20}$ o'rinli bo'lsa, $20 \cdot 19 \cdot C_0 - 19 \cdot 18 \cdot C_1 + 18 \cdot 17 \cdot C_2 - \dots - 3 \cdot 2 \cdot C_{17} + 2C_{18}$ ni hisoblang.

- A) 10200
- B) 10250
- C) 10260
- D) 12000

23. Hisoblang:

$$\int_{-1}^1 \frac{dx}{x^4 + x^2}.$$

- A) $2 - \frac{\pi}{2}$
- B) $-2 - \frac{\pi}{2}$
- C) $-2 + \frac{\pi}{2}$
- D) $2 + \frac{\pi}{2}$

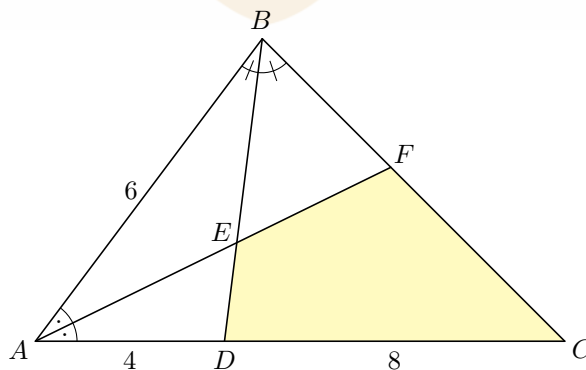
24. ABC to'g'ri burchakli uchburchakning, B to'g'ri burchak bissektrisasi AC gipotenuzasini $\sqrt{5}$ va $2\sqrt{5}$ kesmalarga ajratsa, uchburchak yuzini toping.

- A) 12
- B) 9
- C) 18
- D) 15

25. Aylananing 8 ga teng vatari 90° li yoyni tortib turadi. Aylana radiusini toping.

- A) 8
- B) 4
- C) $4\sqrt{2}$
- D) $8\sqrt{2}$

26. Chizmaga asosan bo'yalgan soha yuzini toping.

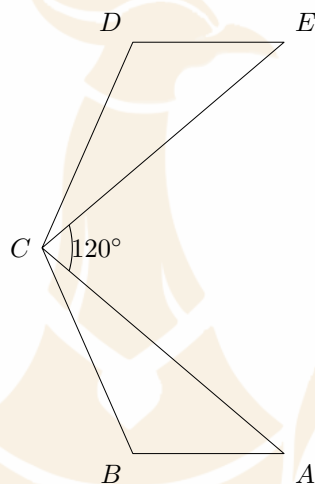


- A) $\frac{24}{5}$
- B) $4\sqrt{5}$
- C) $5\sqrt{5}$
- D) $\frac{24\sqrt{5}}{5}$

27. Trapetsiya asoslari 27 va 7 ga, yon tomonlari 12 va 16 ga teng bo'lsa, trapetsiya yuzini toping.

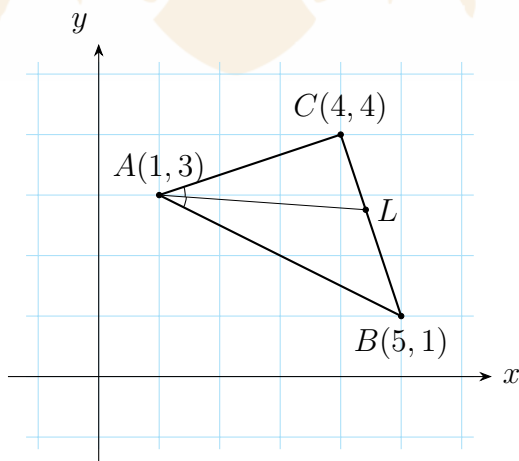
- A) $163\frac{1}{5}$
- B) $162\frac{1}{5}$
- C) $161\frac{1}{5}$
- D) $160\frac{1}{5}$

28. Qavariq muntazam n burchakning kichik dioganallari orasidagi burchak 120° bo'lsa, n ni toping.



- A) 12
- B) 14
- C) 10
- D) 15

29. ABC uchburchakning uchlari koordinatalari berilgan. AL bissektrisasi uzunligini toping.



A) $\sqrt{40 - 10\sqrt{2}}$

B) $\sqrt{40 - 20\sqrt{2}}$

C) $\sqrt{30 - 15\sqrt{2}}$

D) $\sqrt{30 - 10\sqrt{2}}$

30. To'la sirti asosining yuzidan 3 marta katta bo'lgan muntazam to'g'ri to'rtburchakli piramidaning qirrasi asos tekisligi bilan hosil qilgan burchakning tangensini toping.

A) $\sqrt{3}$

B) $\frac{1}{\sqrt{3}}$

C) $\frac{\sqrt{6}}{2}$

D) $\frac{\sqrt{6}}{4}$

31. $A = \{60 \text{ ni natural bo'luvchilari}\}$, $B = \{48 \text{ ni natural bo'luvchilari}\}$ bo'lsa, $A \cap B$ ning qism to'plamlari sonini toping.

A) 32

B) 64

C) 128

D) 16

32. Raqamlari yig'indisi 7 ga teng bo'lsan uch xonali sonlar qog'ozchalarga yozilgan. Bulardan tanlangan son juft son bo'lish ehtimolligini toping.

A) $\frac{1}{2}$

B) $\frac{2}{3}$

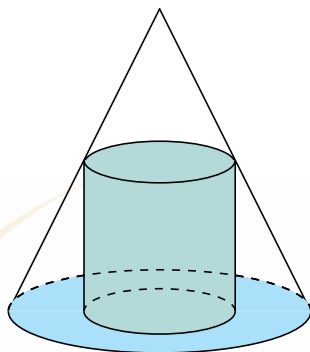
C) $\frac{4}{7}$

D) $\frac{3}{4}$

Topshiriqlar (33–35) va javob variant (A–F) larni o‘zaro moslashtiring.

Javoblar.

Konusga eng katta hajmli silindr ichki chizilgan.



- A) $\frac{1}{3}$
- B) $\frac{2}{3}$
- C) 1
- D) $\frac{3}{4}$
- E) $\frac{4}{9}$
- F) $\frac{2}{9}$

33. $\frac{r_{silindr}}{r_{konus}}$ ni toping.

34. $\frac{h_{silindr}}{h_{konus}}$ ni toping.

35. $\frac{V_{silindr}}{V_{konus}}$ ni toping.

36. Tenglamalar sistemasi berilgan:

$$\begin{cases} 26x^2 + 42xy + 17y^2 = 10 \\ 10x^2 + 18xy + 8y^2 = 6 \end{cases}.$$

a) Tenglamalar sistemasi nechta yechimga ega.

Javob: _____

b) Tenglamalar sistemasining yechimlari uchun $x + y$ ning eng katta qiymatini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

37. Tenglamani yeching:

$$\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3.$$

a) Tenglamaning eng kichik musbat yechimini toping.

Javob: _____

b) Tenglamaning $[-\pi; \pi]$ oraliqdagi ildizlari soni nechta?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

38. $f(x) = \frac{ax+b}{cx+d}$ berilgan. $f(x)$ funksiya Ox o'qini $(3;0)$ nuqtada kesib o'tadi va qiymatlar sohasi $(-\infty;2) \cup (2;\infty)$. $F(x)$ funksiya $f(x)$ ga teskari funksiya bo'lib, $F(x) = f(x)$ shart bajariladi.

a) $f(x)$ ning aniqlanish sohasiga tegishli bo'lmagan sonni toping.

Javob: _____

b) $F(6)$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

39. Bizga $f(x) = \frac{x^2}{8} - \frac{3x}{2}$ parabola va $x^2 + y^2 - 18x - 12y + 97 = 0$ aylana tenglamasi berilgan.

a) Parabola va aylana orasidagi eng yaqin masofani toping.

Javob: _____

b) Aylana markazi va parabola uchi orasidagi masofani toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

40. $f(x) = -2x^2 + 4x$ va $g(x) = x^2 + px + q$ funksiyalar kesishishidan hosil bo'lgan soha yuzi $\frac{32}{27}$ ga teng. $x = \frac{4}{3}$ to'g'ri chiziq chegaralangan soha yuzini teng ikkiga ajratadi.

a) Funksiyalarning kesishgan nuqtalari absissalari yig'indisini toping.

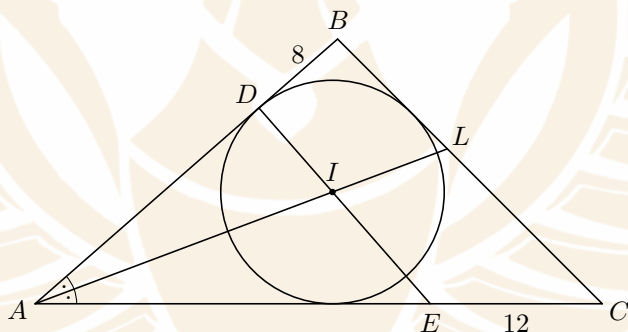
Javob: _____

b) $p \cdot q$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaraqasiga ko'chirib yozing.

41. ABC uchburchakka ichki aylana chizilgan. AL bissektrisa. AB tomonning D nuqtasi aylana urinadi. D nuqtadan BC ga parallel DE chiziq o'tkazilgan. $BC = 30$, $BD = 8$ va $EC = 12$ ga teng.



a) CL ni toping.

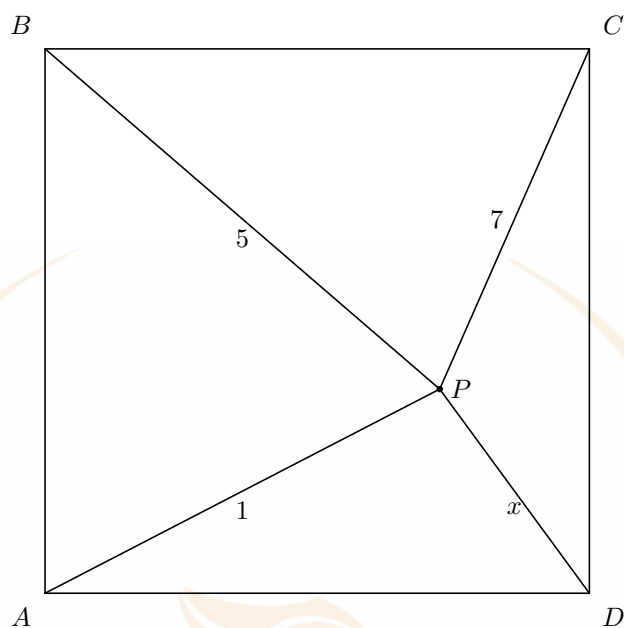
Javob: _____

b) $|AB - AC|$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaraqasiga ko'chirib yozing.

42. $ABCD$ kvadratning ichidan P nuqta olingan. $AP = 1$, $BP = 5$, $PC = 7$ bo'lsin.



a) PD ni toping.

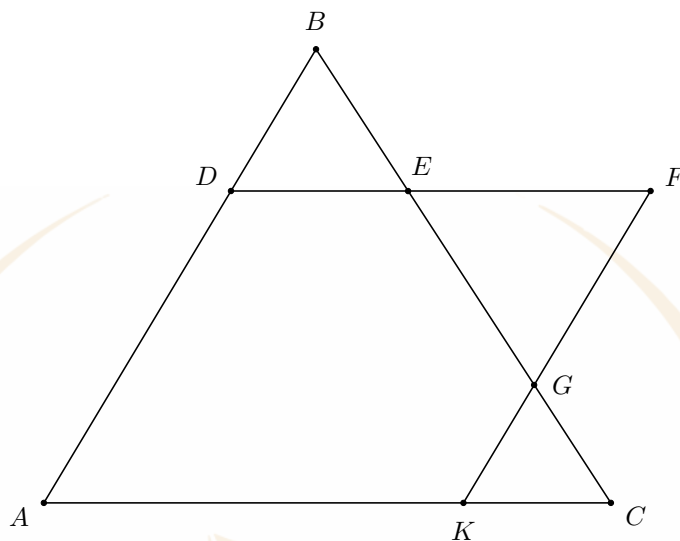
Javob: _____

b) Kvadrat yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

43. Chizmada berilgan $ADFK$ parallelogram va $EG : GC = 1 : 2$, $DE : EF = 2 : 3$ bo'lsin. Chizmaga qarang.



- a) $\frac{EF}{BC}$ ni toping.

Javob: _____

- b) $S_{ABC} = 242$ bo'lsa, S_{ADFK} ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

44. AB diametrli radiusi 3 ga teng sferadan C va D nuqtalar $ABCD$ piramidaning eng katta hajmli bo'ladigan qilib olingan. E nuqta CD ning o'rtasi.

a) Piramida hajmini toping.

Javob: _____

b) AE chiziq va asos tekisligi orasidagi burchak sinusini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqarajasiga ko'chirib yozing.

45. Silindr shaklidagi yog'och hajmi $V = 162\pi \text{ cm}^3$ va yon sirti $S = 72\pi \text{ cm}^2$ ga teng. Duradgor uni yo'nib shar shakliga keltirdi. Sharining hajmi eng katta bo'lsin ($\pi = 3$ deb oling.)

a) Kesib olingan shar sirtining yuzini toping.

Javob: _____

b) G'o'laning chiqindiga chiqqan qismining hajmini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqarajasiga ko'chirib yozing.